BOARD OF DIRECTORS REPORT

Evaluation of Project A (Emerging Markets Expansion) and Project B (Disruptive Technology R&D)

To: Board of Directors, Tiny-Rod Hit Inc.

From: Senior Finance Manager, Corporate Finance

Date: January 2025

Subject: Directional NPV/IRR assessment, project recommendation, risk plan and allocation of $100 million

Decision sought: Approval of the recommended project and, should the Board elect to pursue both, the capital allocation framework in Section 7

1. Executive Summary

The Board asked for a high-level financial and strategic assessment of two growth investments for the upcoming fiscal year: **Project A**, a $50 million expansion into emerging markets with a new product line, and **Project B**, a $40 million R&D programme for a disruptive technology in our existing market. The figures below are **directional estimates** built from the information in the Board reference file plus clearly stated assumptions (Section 2). They are meant to rank the options and shape how we structure them. They are not point valuations.

Indicator (12% risk-adjusted hurdle)	Project A: Emerging Markets	Project B: Disruptive R&D
Capital required	$50M	$40M
NPV, base / expected	≈ +$15M base; ≈ +$18M probability-weighted	≈ +$85M if successful; ≈ +$47M expected (after 30% failure risk)
NPV range across scenarios	≈ −$31M to +$73M	≈ −$40M (failure) to +$250M (strong success)
IRR	≈ 15–16%	≈ 27% if successful; ≈ 22% on expected cash flows
Discounted value per $1 invested (NPV ÷ capital)	≈ $0.35	≈ $1.20
Payback (undiscounted)	≈ 9 years	≈ 6–7 years if successful; never if it fails
Key break-even	NPV ≈ 0 if hurdle rises to ~14% or local currency falls ~3% a year	Expected NPV stays positive unless failure probability exceeds ~68%

Recommendation: Approve **Project B**, funded through a **stage-gated release of the $40M** ($15M / $15M / $10M tied to technical and commercial milestones). Project B creates roughly 2.5–3 times more expected value than Project A, uses less capital and builds on our existing expertise and R&D team. Its main risks are largely within our control, and staged funding limits the loss in most failure cases to about $15M. Project A is value-accretive but marginal once emerging-market risk is priced in. It also depends on capabilities we do not yet have.

If the Board chooses to pursue both projects, we propose: **Project B $40M (40%)**, **Project A $40M (40%)** through a phased, partner-led entry, **$10M (10%) ring-fenced contingency** and a **$10M (10%) strategic liquidity reserve**. On a directional basis this structure lifts the portfolio expected NPV from about +$65M to about +$78M. It also roughly halves the worst-case loss (from about −$71M to about −$38M), and only $40M of cash leaves the balance sheet at launch.

2. Context, Scope and Key Assumptions

2.1 Financial position and decision context

Tiny-Rod Hit Inc. is consistently profitable, has a healthy debt-to-equity ratio and a corporate WACC of about 9%. The reference file shows **$100 million of available cash as of May 2025**. This paper is prepared in January 2025, so we assume the cash is available to deploy from **May 2025**, which is the funding window for the upcoming fiscal year. February to April 2025 is used for the due diligence and gate design described here. Funding both projects in full ($90M) would commit 90% of available cash. How much flexibility we keep is therefore a central part of the allocation question.

2.2 Modelling approach

- **Horizon and valuation:** 10-year explicit free cash flow (FCF) forecast plus a conservative terminal value. FCF margins are stated after tax, maintenance capex and working capital.
- **Discount rates:** The 9% corporate WACC reflects the average risk of our existing business. Both projects are riskier than that, so we apply a **12% project hurdle rate** (9% + 3% premium). For Project A, the premium covers country, currency and execution risk. For Project B, it covers commercial and market risk. Project B's **technical failure risk is modelled separately as a 30% probability of zero revenue**, so it is not counted twice in the discount rate.
- **Risk treatment:** Project A uses bear/base/bull scenarios weighted 25/50/25. Project B uses a decision tree: failure 30%; success split into moderate (25%), base (35%) and strong (10%).
- **Data gap:** The reference file gives growth rates but not starting revenue or margins. The base levels below are management assumptions for a business of our size and should be replaced with bottom-up business cases before final investment approval.

Parameter	Project A: Emerging Markets	Project B: Disruptive R&D
Up-front investment	$50M at Year 0	$40M at Year 0 (lump-sum base case; staged variant tested)
Revenue start (assumed)	Year 1 revenue $20M	Year 1 = R&D/pilot (nil); launch Year 2 at $10M if successful
Revenue growth	20% a year in Years 2–5; 5% a year in Years 6–10 (per reference file)	55% a year in Years 3–5 ("over 50%"); then 35% / 25% / 15% penetration; 8% in Years 9–10
FCF margin	Ramps from 2% to 18% by Year 5 (start-up infrastructure, partner costs)	–10% at launch, rising to 28% at scale by Year 6 (technology-led pricing)
Terminal growth after Year 10	3% (below the 5% stabilised rate, for prudence)	2% (reflects the risk of technological obsolescence)
Discount rate	12% (WACC 9% + 3% emerging-market premium)	12% (WACC 9% + 3% commercial premium) + 30% explicit failure probability
Failure / downside	Bear case: 10% growth, 13% margin, 14% discount rate, 4% annual FX depreciation	Failure: no revenue, full $40M lost, no salvage value assumed (conservative)

3. Project A: Emerging Markets Expansion

3.1 Directional valuation

In the base case, revenue grows from about $20M in Year 1 to about $42M by Year 5 and about $53M by Year 10. FCF builds slowly while infrastructure and partnerships mature, from under $1M in Year 1 to about $9–10M a year by Year 10 (Appendix A).

Scenario (probability)	Key drivers	NPV @ risk-adj. rate	IRR	Payback
Bear (25%)	10% growth, 13% steady margin, 4% annual FX depreciation, 14% discount rate	≈ −$31M	≈ 1%	> 10 yrs
Base (50%)	20% growth Years 2–5, then 5%; 18% steady margin; 12% discount rate	≈ +$15M	≈ 15–16%	≈ 9 yrs
Bull (25%)	$24M Year 1 revenue, 25% growth, 21% margin; 11% discount rate	≈ +$73M	≈ 24%	≈ 6.5 yrs
Probability-weighted		**≈ +$18M**		

3.2 Implications

- **Positive but thin value once risk is priced in.** The base-case IRR of about 15–16% clears the 12% hurdle by only 3–4 points. Discounted at the 9% corporate WACC, the project looks far better (NPV about +$54M), but that would understate the political, regulatory and currency risks the Board has identified. We do not consider 9% the right test for this project.
- **Most of the value comes late.** About 54% of the value comes from the terminal value beyond Year 10. Over the 10-year explicit period alone, NPV is about −$21M. Shareholders therefore depend on the business surviving and staying profitable in markets where we have limited experience.
- **High sensitivity to factors outside our control.** NPV falls to about zero if the required return rises to about 14% (for example, after a sovereign downgrade), or if the local currency loses about 3% a year against the USD. Neither can be managed through operational excellence alone.
- **Strategic value not captured in the NPV.** The project diversifies revenue by geography and customer segment and creates a platform for future products in the region. It also carries an option to expand into neighbouring markets. These qualitative points matter in the portfolio discussion in Section 7.

4. Project B: Disruptive Technology R&D

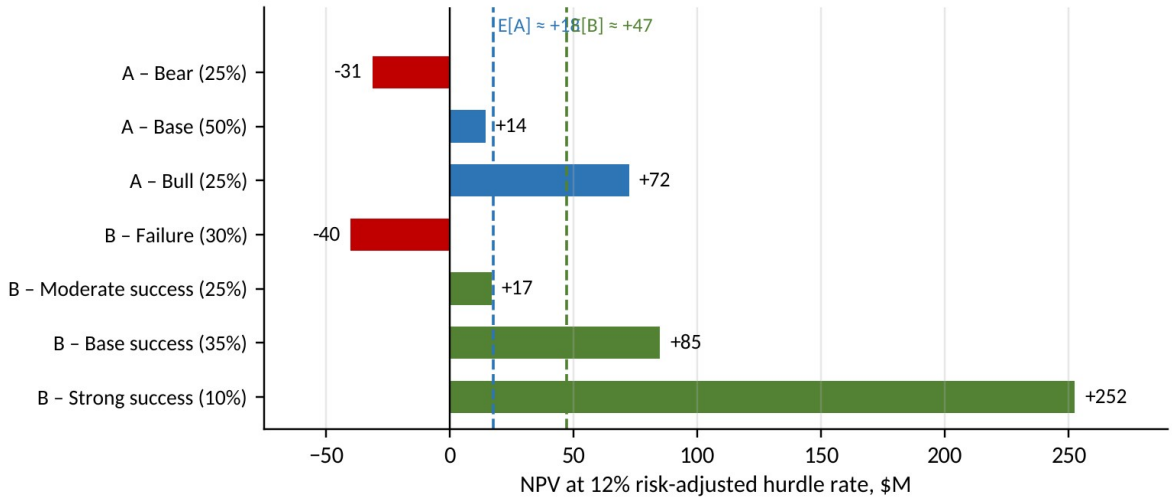
4.1 Directional valuation (decision-tree view)

Project B has a binary outcome. With 30% probability the R&D fails and the $40M is lost. With 70% probability the technology reaches the market in Year 2 and grows by more than 50% a year for three years, followed by rapid market penetration. In the base success case, revenue reaches about $37M by Year 5 and about $84M by Year 10, with FCF of about $24M a year at maturity. The project values below follow from that.

Outcome (probability)	Description	NPV @ 12%	IRR	Payback
Failure (30%)	No significant revenue; full write-off of the $40M	≈ −$40M	n/a	Never
Moderate success (25%)	$8M launch revenue, 50% growth, 22% margin at scale	≈ +$17M	≈ 16–17%	≈ 9 yrs
Base success (35%)	$10M launch revenue, 55% growth, 28% margin at scale	≈ +$85M	≈ 27%	≈ 6.5 yrs
Strong success (10%)	$15M launch revenue, 60–70% growth, 32% margin at scale	≈ +$250M	≈ 41%	≈ 5 yrs
Expected value	**Probability-weighted across all outcomes**	**≈ +$47M**	**≈ 22%***	**≈ 7.5 yrs**

**IRR of the probability-weighted cash flows. For a binary project, IRR is less meaningful than expected NPV and is shown for reference only.*

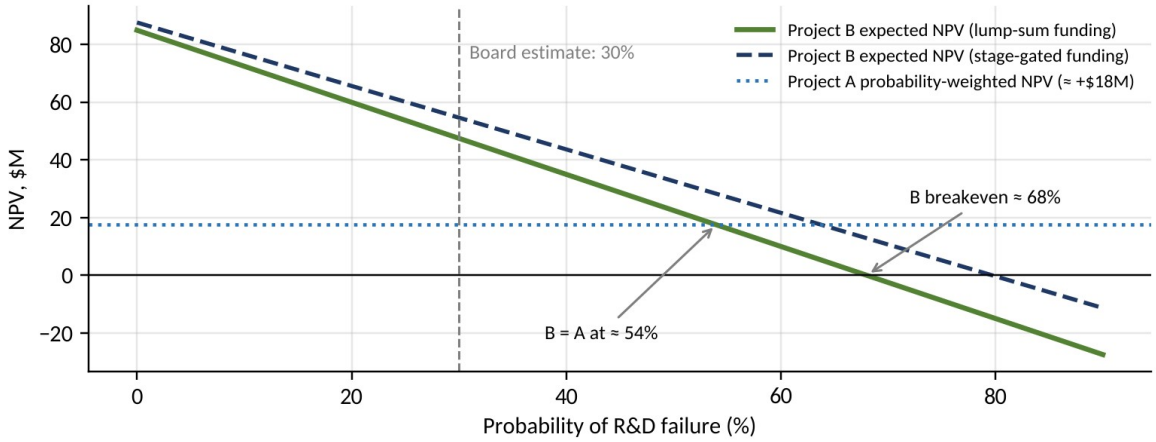
Figure 1: NPV by scenario at the 12% risk-adjusted hurdle rate ($M). Dashed lines show probability-weighted NPV.



4.2 Implications

- **High expected value despite the risk of failure.** After deducting a 30% chance of losing the entire $40M, the expected NPV is still about +$47M, or about $1.20 of value per dollar invested, compared with about $0.35 for Project A.
- **Expected value holds up well against the main uncertainty.** As Figure 2 shows, expected NPV stays positive unless the failure probability exceeds about **68%**, and Project B beats Project A unless the failure probability exceeds about **54%**. Either would require the 30% estimate to be badly wrong. The ranking also holds across discount rates: at 15%, Project B's expected NPV (about +$19M) still exceeds Project A's base case at 12%.
- **Wide range of outcomes; IRR overstates the success case.** The success IRR of about 27% does not reflect the 30% chance of total loss. The Board should judge the project on expected NPV and maximum loss, not on the headline IRR. Value also depends on the terminal value (about 62%), so technological obsolescence is a real risk to the success case.
- **Staged funding adds value.** Releasing the $40M in tranches against milestones, rather than up front, raises expected NPV to about **+$55M**. It also limits the loss to about **$15M** in roughly two-thirds of failure cases, because most failures should surface at the first technical gate.
- **Cost of not investing.** The NPV excludes the defensive value of the project. If a competitor brings a disruptive technology to our core market first, the cash flows of our existing business are at risk. Project B protects the business that funds every other initiative.

Figure 2: Sensitivity of Project B expected NPV to the probability of R&D failure, compared with Project A.



5. Comparative Assessment and Recommendation

5.1 Quantitative comparison

Figure 3: Cumulative undiscounted cash flow profiles, Years 0–10 (excluding terminal value), $M.

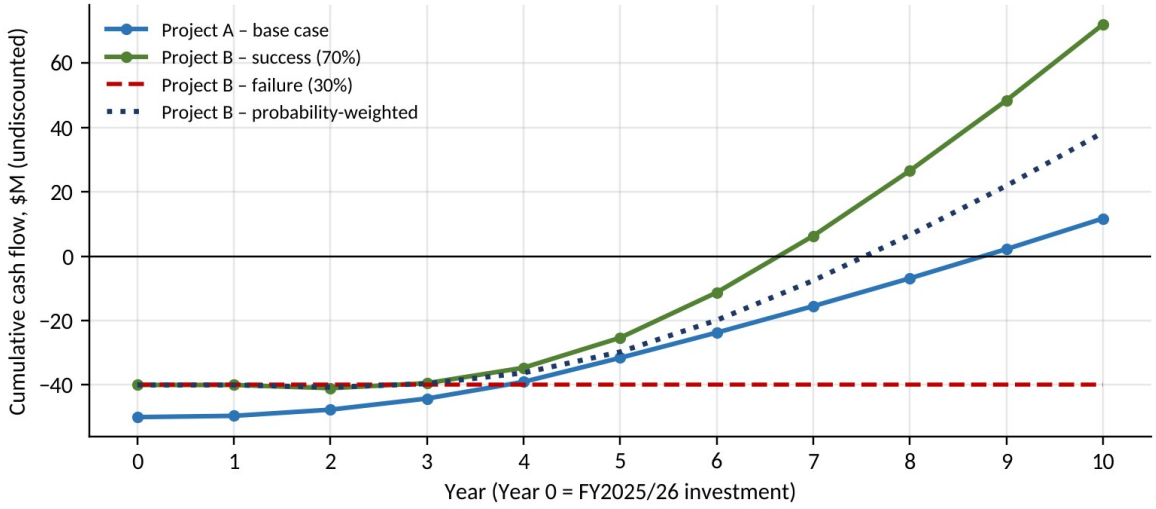


Figure 3 shows the different risk profiles. Project A has a long, shallow J-curve and does not recover its cash until about Year 9. Project B either loses $40M outright or recovers its investment by about Year 6–7 and then generates cash much faster. Weighted by probability, Project B recovers its cash earlier than Project A and keeps $10M more cash on the balance sheet at the outset.

5.2 Qualitative and strategic scorecard

Criterion (weight)	Project A	Project B	Commentary
Strategic alignment (20%)	4	5	A meets the diversification goal; B secures market leadership in our core market and could redefine the industry
Capability fit / execution confidence (20%)	2	5	A requires new infrastructure, local partners and market know-how we lack; B uses our proven R&D team and core-market expertise
Risk controllability (15%)	2	4	A's key risks (geopolitics, regulation, FX) are external; B's (technology, IP) can be managed internally and through staged funding
Diversification benefit (15%)	5	2	A adds new geographies and customer segments; B increases concentration in the existing market
Long-term value and upside (15%)	4	5	Both offer growth platforms; B's success case changes the company's value
Capital efficiency and time to value (15%)	2	4	B needs less capital, has higher value per dollar and pays back faster if successful
Weighted score (out of 5)	**3.2**	**4.3**	**Project B preferred on a risk-adjusted strategic basis**

5.3 Recommendation

We recommend that the Board approve Project B, with the $40M released in stages against milestones. Five reasons support this:

1. **Value:** Expected NPV of about +$47M (about +$55M with staged funding), against about +$18M for Project A, using $10M less capital.
2. **Robustness:** The case holds unless the failure probability more than doubles from the 30% estimate. Project A's value disappears with a 2-point rise in the required return or a modest fall in the local currency.
3. **Capability:** Project B relies on capabilities we already have: a strong internal R&D team, deep customer relationships and knowledge of our core market. Project A depends on capabilities we would have to build or buy.
4. **Strategic defence:** Leading the disruption in our core market protects the profit engine that pays for future diversification, including Project A at a later stage.
5. **Limited downside:** Staged funding turns a $40M all-or-nothing bet into a series of options, each exercised only once new evidence supports it.

We acknowledge that Project B increases concentration in our existing market. Project A's diversification benefits are real. Section 7 sets out how both could be pursued responsibly if the Board prefers.

6. Project B: Top Three Risks, Mitigation and Contingency Plans

The three risks below have the greatest combined effect on value and likelihood. Each has a financial and an operational dimension. Each one has key risk indicators (KRIs) that trigger escalation to the Investment Committee.

Risk 1: R&D failure and budget overrun (financial)

Assessment	Mitigation strategies	Contingency plan if the risk materialises
High likelihood (30% failure; R&D programmes often overrun by 15–25%). Impact: write-off of up to $40M, an earnings charge and lost opportunity. **KRIs:** milestone slippage of more than one quarter; spend more than 10% above plan; prototype performance below target.	• **Stage-gated funding:** $15M Phase 1 (proof of concept) → $15M Gate 1 (~month 12: prototype meets at least 80% of target specification) → $10M Gate 2 (~months 18–24: pilot customer validation and unit economics within 15% of plan). • **Kill criteria agreed in advance** and assessed by an independent technical review panel (internal CTO office plus external experts) so that past spending does not drive the decision. • **Run alternative technical approaches in parallel** for the highest-risk components during Phase 1, then narrow to one at Gate 1. • **Cost discipline:** milestone-based payments to vendors and contractors, monthly tracking of spend against progress, and use of R&D tax credits and grants where available.	• If Gate 1 fails: stop the programme, cap the loss at about $15M and release the remaining $25M to the reserve or Project A. • Recover value from the work: license or sell patents and reuse the technology in existing product lines. • If overruns exceed 15%: draw on the $5M ring-fenced contingency only with CFO and Investment Committee approval and a revised scope; otherwise narrow the target application.

Risk 2: Competitive response and technological obsolescence (financial / commercial)

Assessment	Mitigation strategies	Contingency plan if the risk materialises
Medium–high likelihood. Established players may copy us quickly, cut prices or bundle; a different technology could leapfrog ours. Impact: lower growth and margins and a shorter economic life (62% of success-case value sits beyond Year 10). **KRIs:** competitor product or patent announcements; pilot win rate below 50%; price erosion above 10% in pilots.	• **Speed to market:** a minimum viable product with 3–5 existing customers as design partners, with letters of intent or pre-orders before Gate 2. • **Customer lock-in:** integrate the technology with our existing installed base, services and data so that switching costs exceed the price of competing offers. • **Technology and competitor monitoring:** quarterly scans of patents, start-ups and competitors; modular architecture so that components can be replaced as technology changes. • **Value-based pricing** with multi-year contracts for anchor customers to secure early revenue.	• Use the acceleration contingency to bring launch forward, or form a partnership or make a bolt-on acquisition to close capability gaps. • Reposition to a premium niche where performance matters more than price, or bundle with core products to protect share. • If a rival standard wins: license our IP to the market leader to earn royalties instead of competing directly.

Risk 3: IP protection and loss of key R&D talent (operational)

Assessment	Mitigation strategies	Contingency plan if the risk materialises
Medium likelihood, high impact. The value of a disruptive technology depends on defensible IP and on a small number of key engineers. Infringement claims, leaks or departures could remove our competitive advantage. **KRIs:** departure of any designated key person; regretted attrition above 10% a year; third-party patent filings in the same area.	• **IP strategy from day one:** early provisional patent filings, a freedom-to-operate review before Gate 2, trade-secret controls (restricted access, data-loss prevention) and invention-assignment clauses in all contracts. • **Talent retention:** retention awards and long-term incentives tied to Gate 2 and commercial launch; key-person mapping, documented knowledge and identified successors for critical roles. • **Partner safeguards:** strict confidentiality and IP-ownership terms with universities, vendors and pilot customers.	• Draw from the contingency (around $1–2M) for IP defence; seek cross-licences or settlements where litigation is uneconomic. • Publish non-core inventions defensively so that others cannot patent them. • Line up research institutes and specialist contractors in advance to backfill expertise quickly; consider hiring whole teams through small acquisitions.

7. Pursuing Both Projects: Allocation of the $100 Million

As the Board requested, this section assumes both projects are viable. Our financial position (consistent profits, a healthy debt-to-equity ratio and $100M of available cash) can support both, provided capital is released against evidence rather than committed up front.

7.1 Allocation principles

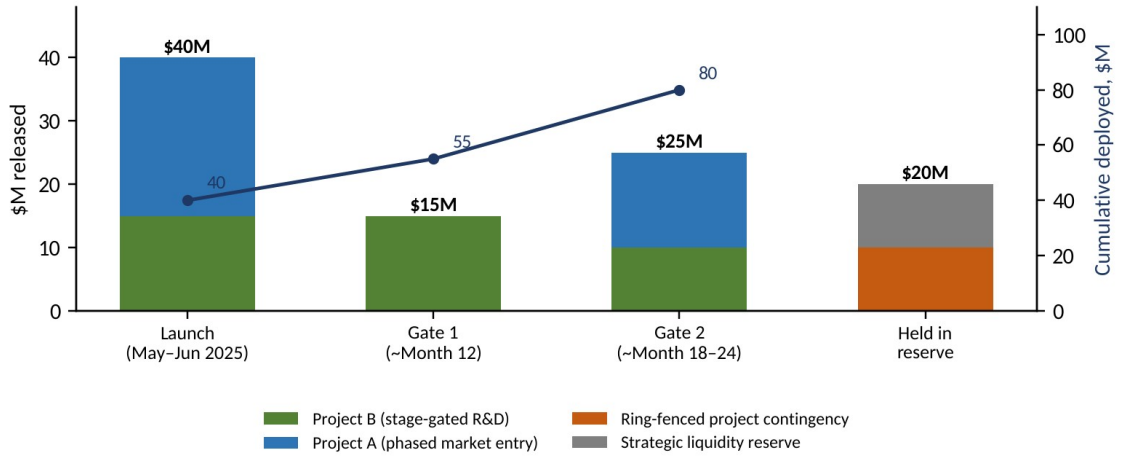
- **Allocate by value and by how each project scales, not equally.** R&D does not scale down well: underfunding Project B increases its chance of failure, so it receives its full $40M request. Market entry can be done in modules: Project A can start in one or two lead markets and expand once there is evidence.
- **Stage capital against evidence.** Each tranche is an option exercised only when a gate is passed, which reduces the cost of being wrong.
- **Combine projects whose risks are unrelated.** Project A's risks (macro conditions, currency, politics) have little in common with Project B's (technology, competitive response), so the portfolio is more resilient than either project alone.
- **Keep liquidity and the credit rating intact.** At least 10% stays unallocated, and our healthy debt-to-equity ratio provides borrowing capacity as a second line of defence.

7.2 Proposed allocation

Allocation	Amount	%	Release schedule	Release conditions
Project B: disruptive R&D (stage-gated)	$40M	40%	$15M at launch (May 2025); $15M at Gate 1 (~month 12); $10M at Gate 2 (~months 18–24)	Prototype meets at least 80% of specification; pilot customer validation; freedom-to-operate clearance; unit economics within 15% of plan
Project A: emerging markets (phased, partner-led)	$40M	40%	$25M Phase 1: 1–2 lead markets; $15M Phase 2 (~month 24): expansion	Phase 1 revenue at least 80% of plan; margin on track; licences secured; partner performing; FX and country-risk indicators within tolerance
Ring-fenced project contingency	$10M	10%	On demand: $5M for Project B (overrun, acceleration, IP defence); $5M for Project A (FX, regulatory delays)	CFO and Investment Committee approval; reported to the Board
Strategic liquidity reserve	$10M	10%	Not allocated	Operating buffer and opportunistic use; Board approval required
Total	**$100M**	**100%**	**Peak cash out at launch: $40M**	

Why Project A receives $40M rather than $50M. We propose an asset-light entry: a local distribution and manufacturing partner provides part of the infrastructure in return for about 10% of project cash flow. This lowers our capital need, brings local market and regulatory knowledge (addressing our lack of experience in these markets), and shares political risk. Directionally, the phased structure lifts Project A's probability-weighted NPV from about +$18M to about +$23M. It also halves the bear-case loss (about –$16M against –$31M), because Phase 2 would not go ahead in a bear scenario.

Figure 4: Capital release schedule. Only $40M is spent at launch; the remainder depends on gates being passed.



7.3 Portfolio effects

Portfolio metric (directional)	Both funded up front ($90M)	Recommended structure
Expected portfolio NPV	≈ +$65M	≈ +$78M
Probability portfolio NPV is negative	≈ 29%	≈ 22%
Worst case (Project A bear and Project B fails)	≈ −$71M	≈ −$38M
Cash spent at launch / still held	$90M / $10M	$40M / $60M
Probability both projects disappoint at once	≈ 7.5% (25% × 30%, risks largely unrelated)	Same probability; loss roughly halved by gates

The joint scenario matrix below shows portfolio NPV ($M) under the recommended structure. Portfolio value is negative in only three of twelve combinations, and even the worst combination costs less than 40% of available cash.

Project A ↓ / Project B →	Failure (30%)	Moderate (25%)	Base (35%)	Strong (10%)
Bear (25%)	≈ −38	≈ +4	≈ +72	≈ +240
Base (50%)	≈ −3	≈ +39	≈ +107	≈ +275
Bull (25%)	≈ +48	≈ +90	≈ +158	≈ +325

7.4 Long-term value creation, diversification and strategic alignment

- **Combined long-term value:** Project B strengthens and extends our core franchise; Project A builds a second growth engine in faster-growing economies. Together they meet both of the Board's stated objectives: market leadership and diversification.
- **Synergy:** Project A's distribution network and local partnerships can become the launch channel for Project B's technology in emerging markets from about Year 4. This would extend Project B's economic life and strengthen Project A's product range. Neither standalone NPV includes this option.
- **Diversification:** Project A spreads revenue across geographies and customer segments. This offsets the extra concentration in our core market that Project B creates and reduces dependence on one market cycle.
- **Cash flow timing:** If successful, Project B produces cash from about Year 3 and pays back by Years 6–7. Project A ramps more slowly but steadily. Early cash from Project B can fund Project A's expansion and later stages without new external financing.

- **Project A safeguards:** political-risk insurance (multilateral or export-credit agency cover), local-currency borrowing and selective FX hedging to offset currency exposure, local regulatory counsel, a staged choice of countries (stable regulatory regimes first), and joint-venture terms with clear exit and buy-out rights.

7.5 Rules for reallocating capital

- **Project B fails Gate 1:** the unspent $25M returns to the reserve and may bring Project A Phase 2 forward, fund other priorities, or be returned to shareholders.
- **Project A misses Phase 1 milestones:** the $15M Phase 2 is withheld. It may be used to scale up Project B commercially (sales, production capacity) if Gate 2 is passed.
- **Both succeed beyond plan:** fund extra scale-up first from operating cash flow, then from debt capacity, keeping the $10M reserve and the debt-to-equity ratio within policy.

8. Governance and Decisions Requested

Governance: The Investment Committee (CEO, CFO, CTO, Head of Strategy) will review monthly KPI and KRI dashboards and decide on releasing contingency funds. The Board will approve each gate or phase tranche at a semi-annual review and receive a post-investment review 24 months after launch.

The Board is asked to:

1. **Approve Project B** with up to $40M of stage-gated funding, releasing the $15M Phase 1 tranche from May 2025.
2. **Note the dual-investment framework** in Section 7. If the Board elects to pursue both projects, approve Project A Phase 1 ($25M), subject to partner and country due diligence, with Phase 2 ($15M) subject to the gate conditions.
3. **Approve the $10M ring-fenced contingency** and keep a **$10M unallocated liquidity reserve**.
4. **Direct management** to prepare detailed bottom-up business cases, gate criteria and partner term sheets by April 2025 to validate the directional estimates in this report.

These estimates rely on management assumptions where the reference file gave no data (starting revenue, margins, timing). They should be read as indicating direction and relative size, not as precise valuations. The ranking of the two projects and the case for staged funding hold across the sensitivities we tested.

Appendix A: Indicative Cash Flow Projections ($M, base cases)

$M	Y0	Y1	Y2	Y3	Y4	Y5	Y6	Y7	Y8	Y9	Y10
A: Revenue		20.0	24.0	28.8	34.6	41.5	43.5	45.7	48.0	50.4	52.9
A: Free cash flow	(50.0)	0.4	1.9	3.5	5.2	7.5	7.8	8.2	8.6	9.1	9.5
B (success): Revenue		0.0	10.0	15.5	24.0	37.2	50.3	62.8	72.3	78.0	84.3
B (success): Free cash flow	(40.0)	0.0	(1.0)	1.6	4.8	9.3	14.1	17.6	20.2	21.9	23.6

Terminal values at Year 10: Project A ≈ $109M (3% growth, 12% discount rate); Project B success ≈ $241M (2% growth, 12% discount rate). In present-value terms these are about 54% and 62% of each project's gross value. Project B failure case: $40M outflow at Year 0 and no further cash flows. Staged variant: $15M at Year 0 and $25M at Year 1, conditional on Gate 1; two-thirds of failures are assumed to be caught at Gate 1. Project A phased variant: $25M at Year 0 (55% of base cash flows); $15M at Year 2 (remaining 45%, from Year 4) only where it adds value; 10% partner share of cash flow.